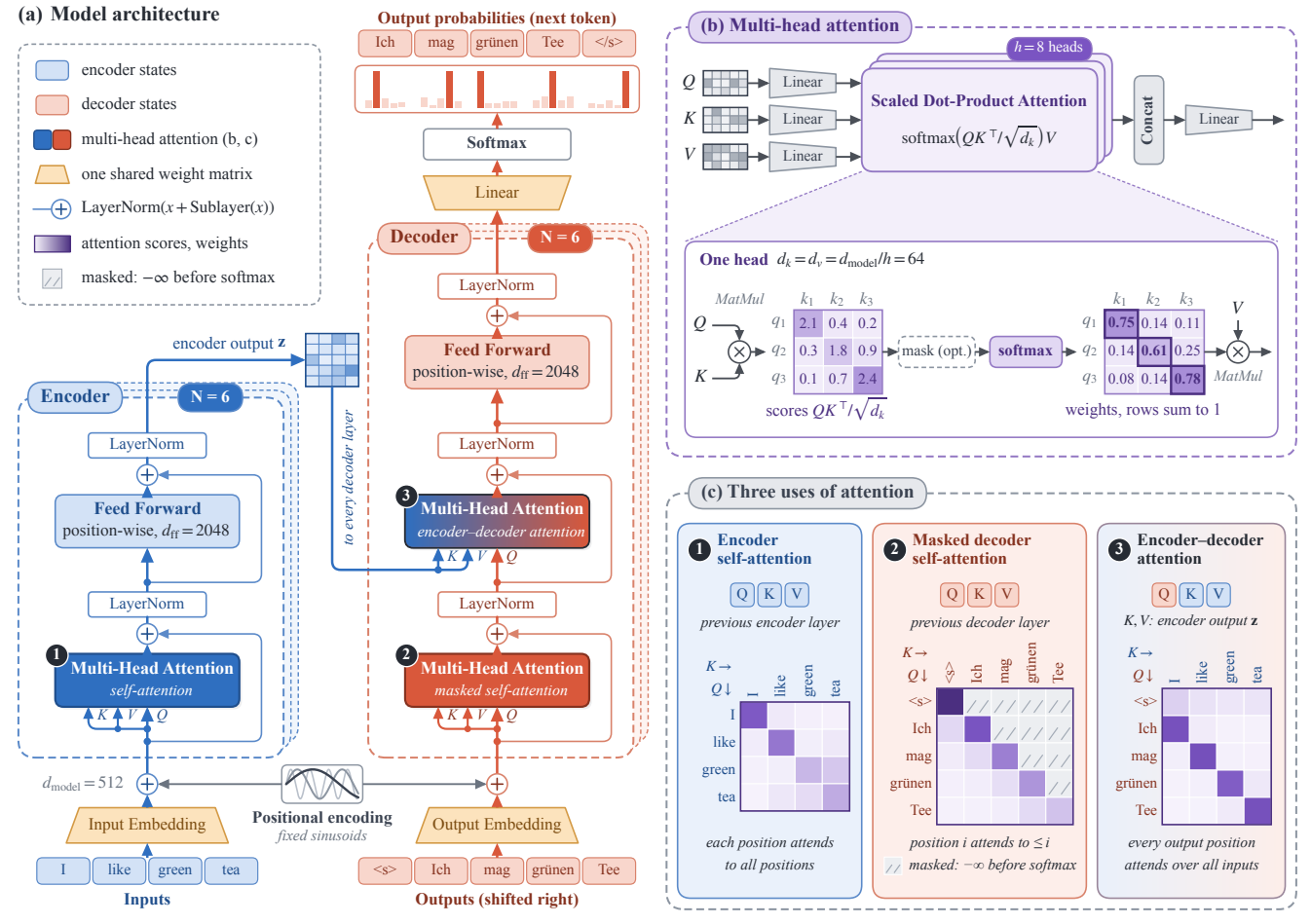

(a) Model architecture
encoder states
decoder states
multi-head attention (b, c)
one shared weight matrix
LayerNorm(x + Sublayer(x))
attention scores, weights
masked: −∞ before softmax
Output probabilities (next token)
Ich
mag
grünen
Tee
</s>
Softmax
Linear
Decoder
N = 6
LayerNorm
Feed Forward
position-wise, $d_{ff} = 2048$
Multi-Head Attention
encoder–decoder attention
Multi-Head Attention
masked self-attention
encoder output z
to every decoder layer
Encoder
N = 6
LayerNorm
Feed Forward
position-wise, $d_{ff} = 2048$
Multi-Head Attention
self-attention
$d_{model} = 512$
Input Embedding
Positional encoding
fixed sinusoids
Output Embedding
I
like
green
tea
Inputs
<s>
Ich
mag
grünen
Tee
Outputs (shifted right)
(b) Multi-head attention
h = 8 heads
Q
K
V
Linear
Scaled Dot-Product Attention
$\mathrm{softmax}(QK^\top/\sqrt{d_k})V$
Concat
Linear
One head $d_k = d_v = d_{model}/h = 64$
MatMul
mask (opt.)
softmax
MatMul
scores $QK^\top/\sqrt{d_k}$
weights, rows sum to 1
(c) Three uses of attention
1 Encoder self-attention
previous encoder layer
each position attends to all positions
2 Masked decoder self-attention
previous decoder layer
position i attends to ≤ i
masked: −∞ before softmax
3 Encoder–decoder attention
K, V: encoder output z
every output position attends over all inputs